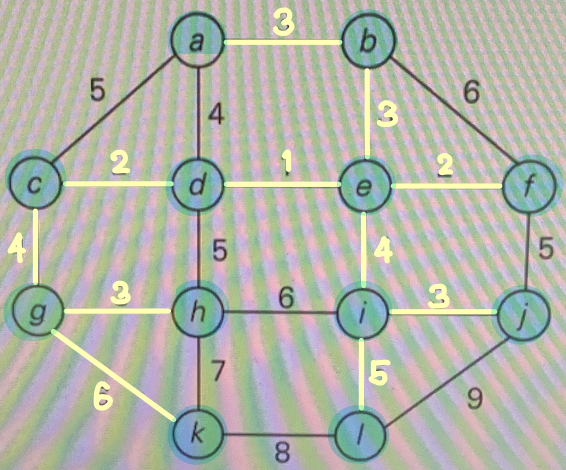
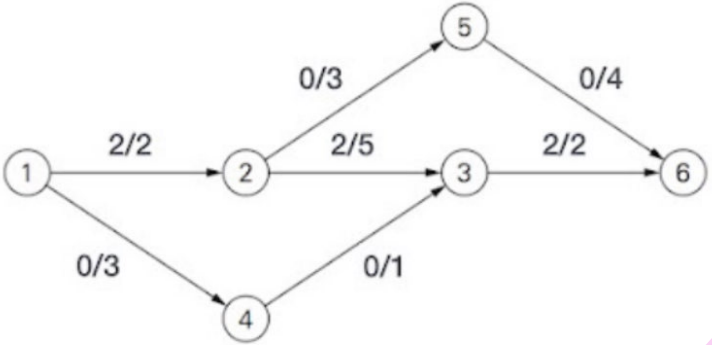
What is the cost of minimum spanning tree of the graph below ? *

2 po

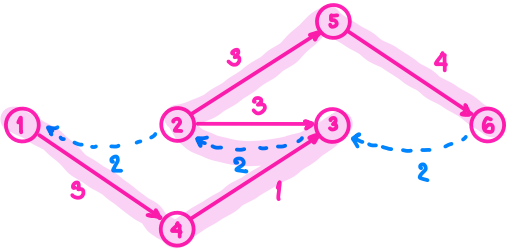


- 28
- 31
- 33
- 36 ✓
- 38
- 41

For the following flow network with source node 1 and destination node 6, which one is a valid augmenting path in the residual graph ? * 1 point



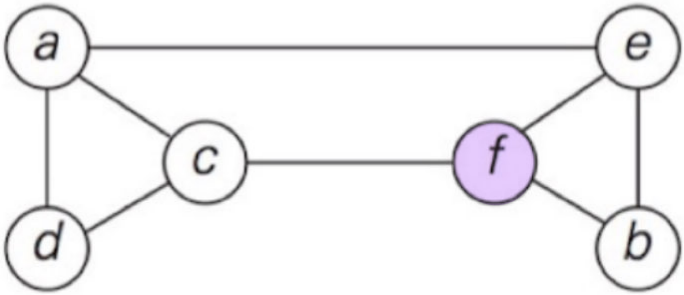
- [ ] 1 -> 2 -> 3 -> 4
- [x] 1 -> 4 -> 3 -> 2 -> 5 -> 6
- [ ] 1-> 2 -> 5 -> 6
- [ ] 1 -> 4 -> 3 -> 6



If we run Depth first search by starting at node A, and then the subsequent nodes are visited alphabetically, what is the order at which node F is visited? * 1 point

A - Z

start



- 2
- 3
- 4
- 5
- 6

4

f_4,

d_3, 1

c_2,

a_1,

Which problem(s) below can be solved directly using a minimum spanning tree algorithm ? * 2 points

multidimensional scaling technique

- [ ] Discovering a cluster of customers with similar characteristics. x MST clustering
- [ ] Finding a minimum cost path between any two nodes in a graph. x shortest path
- [ ] Finding a strongly connected component (SCC) in a graph. x graph traversal
- [x] Designing a transmission network with smallest total link cost. ✓ minimum spanning
- [ ] Finding a largest cost path between any two nodes in a graph. x DAG algorithm

Which abstract data structure is used during an execution of Prim's algorithm to determine the next node to be added to the tree? * 1 point

edges vertices

- ( ) List → O(E + VlogV)
- (•) Priority queue (Min Heap)
- ( ) Dictionary
- ( ) Singly-linked list
- ( ) Binary search tree

An edge in a flow network has the edge capacity of 16 units and a flow of 5 units. What are the capacities of the forward edge and the reverse edge in the residual graph ? * 2 points



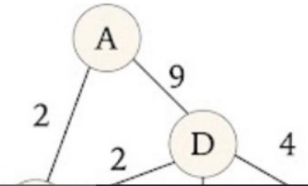
- Forward edge capacity of 5 units and Reverse edge capacity of 5 units
- Forward edge capacity of 16 units and Reverse edge capacity of 11 units
- Forward edge capacity of 11 units and Reverse edge capacity of 5 units
- Forward edge capacity of 11 units and Reverse edge capacity of 16 units
- Forward edge capacity of 5 units and Reverse edge capacity of 11 units

Cycle detection is needed during an execution of Kruskal's algorithm. * 1 point

ensure no loop
cycle detection need!

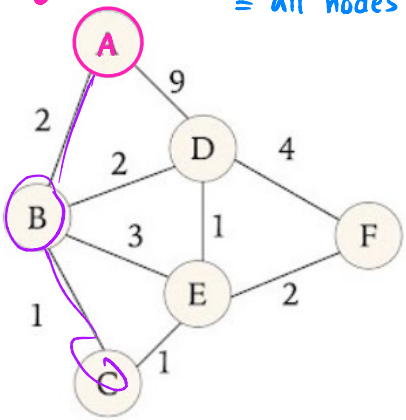
- [x] True
- [ ] False

From the graph below, if we apply Dijkstra's algorithm with node A as the source node, the labels of which nodes will change in the 2nd iteration when the edge relaxation is performed ? * 1 point



From the graph below, if we apply Dijkstra's algorithm with node A as the source node, the labels of which nodes will change in the 2nd iteration when the edge relaxation is performed ? * 1 point

Initialize A = all nodes connected to u



$B_{A,2}$

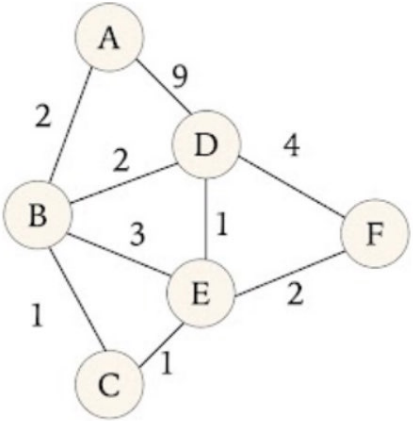
$\subset_1$

E

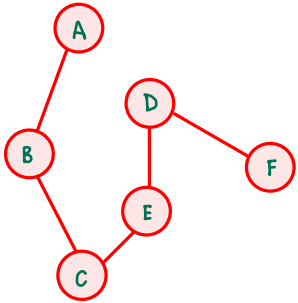
- [ ] A
- [ ] B
- [ ] C
- [x] D
- [ ] E
- [ ] F

This is a required question

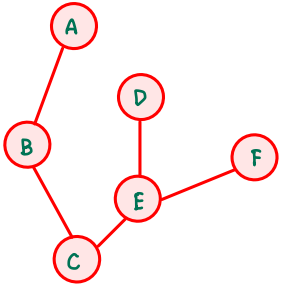
From the graph below, the minimum spanning tree is the same as the shortest path tree rooted at node A. * 2 points



SHORTEST PATH



Minimum Spanning Tree

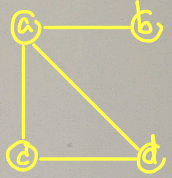


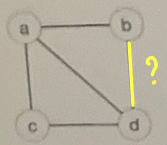
- ○ True
- ● False

Given the adjacency matrix of graph G as shown below, which subgraphs are not spanning trees of G ? * 2 points

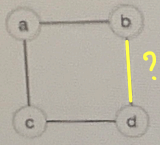
Adjacency matrix of G

	a	b	c	d
a	0	1	1	1
b	1	0	0	0
c	1	0	0	1
d	1	0	1	0

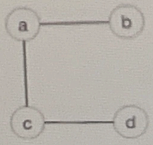




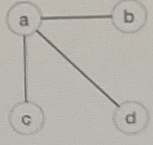
G_1



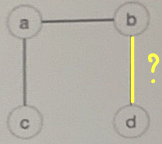
G_2



G_3



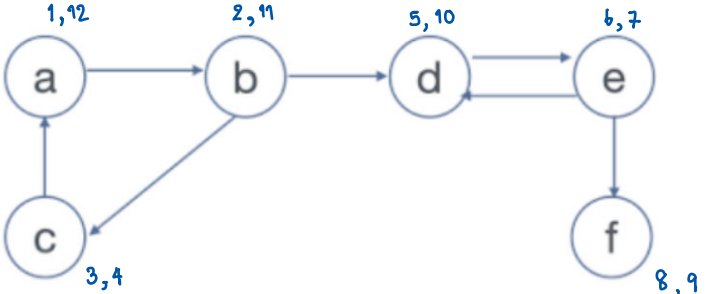
G_4



G_5

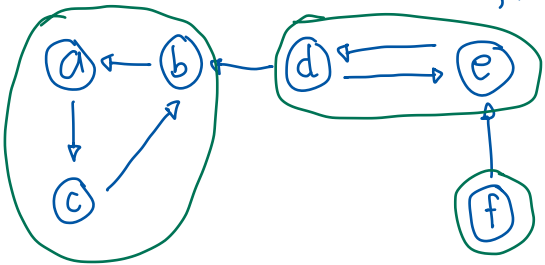
- [x] G1
- [x] G2
- [ ] G3
- [ ] G4
- [x] G5

How many strongly connected components are there in the graph below? * 1 point



a
b
d
e
c
f

- 2
- 3
- 4
- 5
- 6



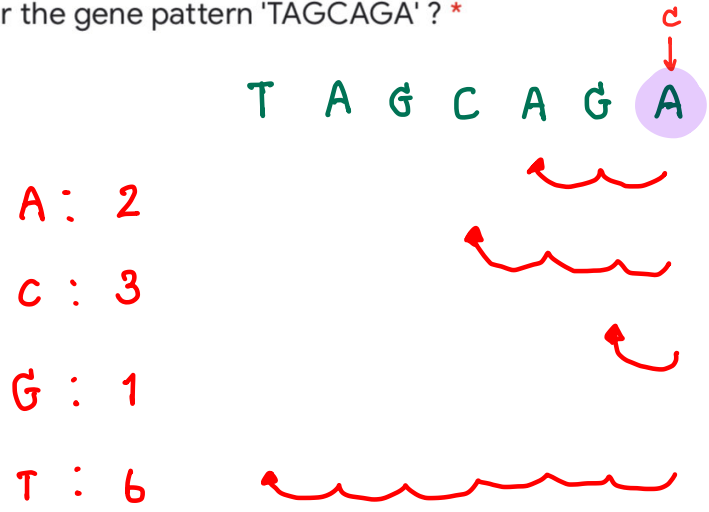
1. a-c-b
2. d-e
3. f

3 SCCs

a-b-c → d-e → f

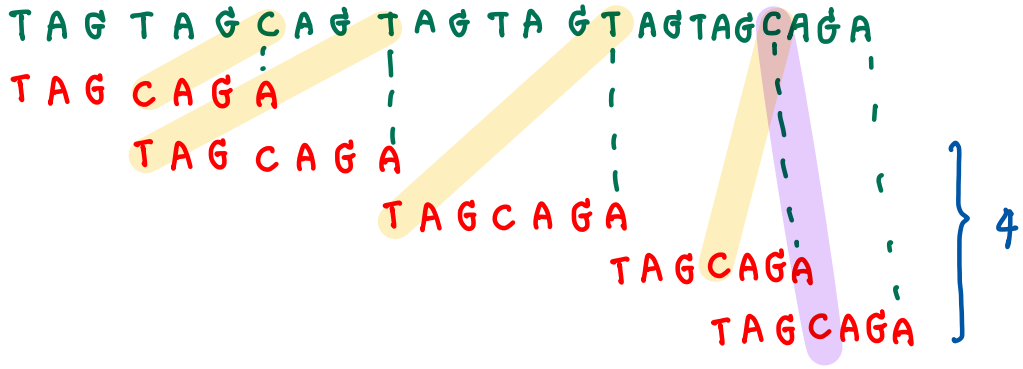
9. Suppose you want to search for a DNA gene pattern in a DNA string by using Horspool's algorithm. If each position in a DNS string is taken from one of the four letters 'A', 'C', 'G', 'T', what are the shift values of these four letters in the shift table for the gene pattern 'TAGCAGA' ? * 1/1

- (●) 2, 3, 1, 6
- ( ) 5, 3, 1, 6
- ( ) 7, 3, 1, 6
- ( ) 3, 4, 2, 6
- ( ) 2, 3, 2, 6

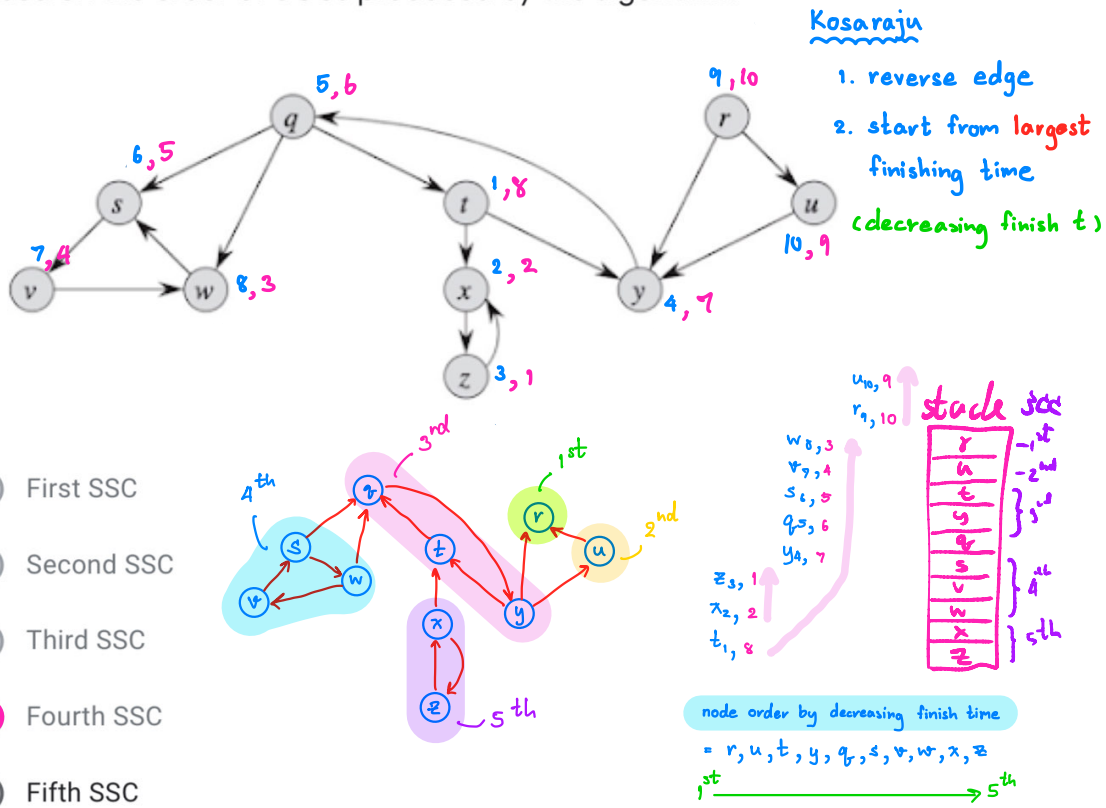


10. (2 points) Suppose you want to search for a DNA gene pattern in a DNA string by using Horspool's algorithm. If each position in a DNS string is taken from one of the four letters 'A', 'C', 'G', 'T', how many pattern shifts does it take to find the pattern 'TAGCAGA' in the DNA string 'TAGTAGCAGTAGTAGTAGTAGCAGA' ? * 2/2

- 2
- 3
- 4
- 5
- 6

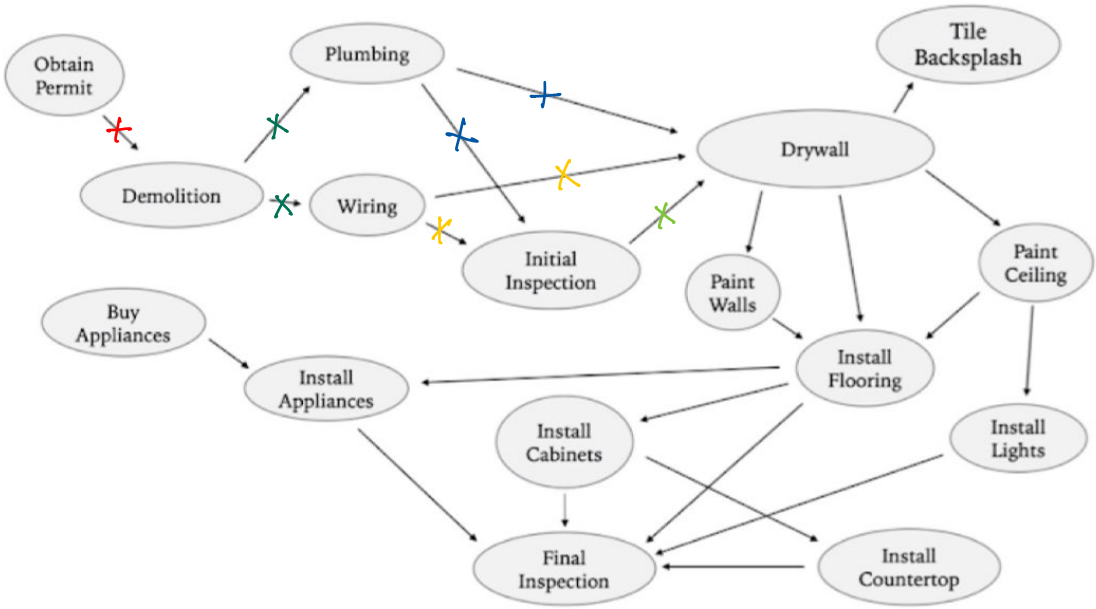


14. (3 points) From the graph below, suppose we run Kosaraju's algorithm starting at node t (break ties alphabetically), node w will be in which SCC based on the order of SCCs produced by the algorithm? * 0/3



- First SSC
- Second SSC
- Third SSC
- Fourth SSC
- Fifth SSC

15. Given the task dependencies of kitchen remodeling project below, which task below cannot be done within the first five orders based on the topological order produced by the source removal algorithm? * 1/1

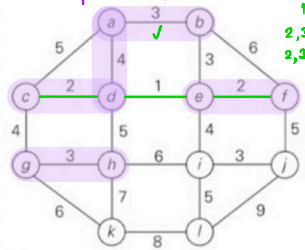


- ○ Plumbing
- ● Drywall
- ○ Buy Appliances
- ○ Obtain Permit
- ○ Wiring
- ○ Initial inspection

ข้อนี้พยายามทำความเข้าใจนะ ข้อสอบจริงมีโอกาส 50% ที่ภาพในโจทย์จะเปลี่ยนนิดนึง แล้วคำตอบมันจะไม่ตรงเฉลย

If Kruskral's algorithm is applied to the graph below, which edges could be added to the spanning tree in the fourth order? * 2 points

node from choice



Edge order

order

1 (d,e) = 1 min

2,3 (c,d) = 2 ✗

2,3 (e,f) = 2 ✗

(a,b) = 3

(b,e) = 3

(g,h) = 3

(i,j) = 3

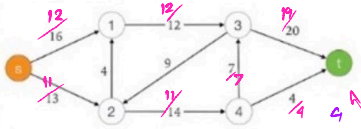
4-7

(a,d) = 4 ✗ max

↳ max node from choice

- [x] (a,b) ✓
- [ ] (c,d) → cannot be 4th order } already use in 2nd-3rd order
- [ ] (e,f) → cannot be 4th order
- [ ] (d,a) → cannot be 4th order → the edge =3 hasn't add yet
- [x] (h,g)

For the following s-t flow network, what is the maximum s-t flow ? * 3 points



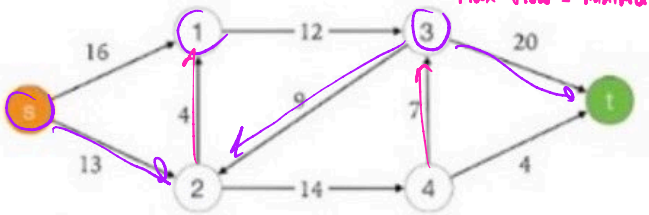
- [ ] 10
- [ ] 17
- [ ] 19
- [x] 23
- [ ] 27

Tips: calculate from t

For the following s-t flow network, what is the minimum cut ? *

2 points

max flow = minimum cut capacity



- ○ {s, 1} 25 = 13+12
- ○ {s, 1, 2} 26 = 12+14
- ○ {s, 1, 3} 42 = 13+9+20
- ● {s, 1, 2, 4} 23 = 12+7+4
- ◉ {s, 1, 3, 4} 46 = 13+9+20+4

1/2 0/2

7/1

Given a list (9, 12, 5, 30, 17, 20, 8, 4), what is the result of Lumoto partition? * 3 points

- (8,4,5), 9, (20,17,30,12)
- (4,8,5), 9, (17,12,30,20)
- (8,4,5), 9, (17,20,30,12)
- (4,5,8), 9, (17,20,12,30)
- (8,4,5), 9, (30,20,17,12)

